

CODE :-

```
import numpy as np
import matplotlib.pyplot as mp

def sigmoid(x):
    a=1/(1+np.exp(-x))
    return a

def tanhh(x):
    return np.tanh(x)

def relu(x):
    return np.maximum(0,x)

x=np.linspace(-10,10,20)
y=sigmoid(x)
y1=tanhh(x)
y2=relu(x)

mp.figure(figsize=(15,10))
mp.title("Activation Function",color="red",fontsize=20)

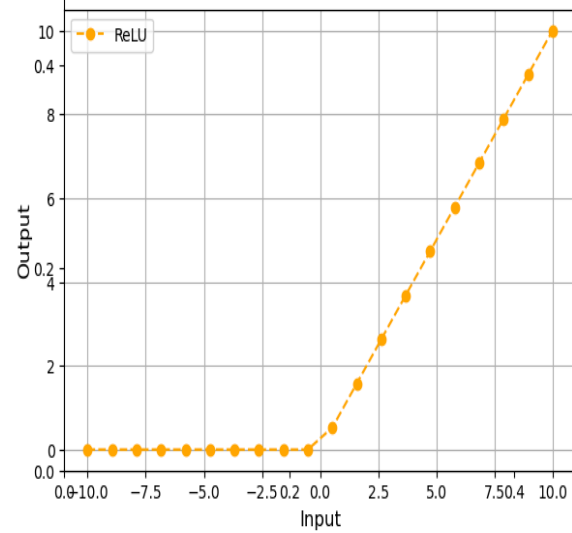
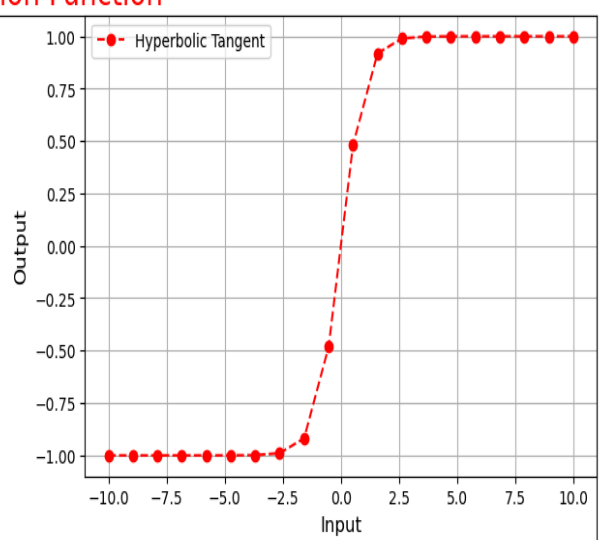
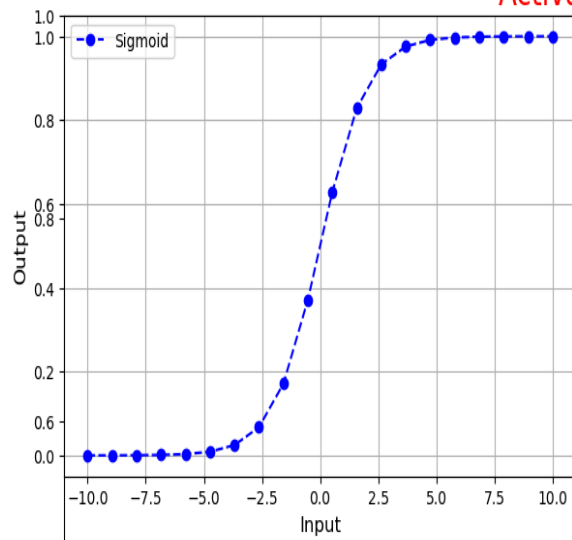
mp.subplot(2,2,1)
mp.plot(x,y,color="blue",marker="o",linestyle="--",label="Sigmoid")
mp.xlabel("Input",fontsize=12)
mp.ylabel("Output",fontsize=12)
mp.legend()
mp.grid()

mp.subplot(2,2,2)
mp.plot(x,y1,color="red",marker="o",linestyle="--",label="Hyperbolic Tangent")
mp.xlabel("Input",fontsize=12)
mp.ylabel("Output",fontsize=12)
mp.legend()
mp.grid()

mp.subplot(2,2,3)
mp.plot(x,y2,color="orange",marker="o",linestyle="--",label="ReLU")
mp.xlabel("Input",fontsize=12)
mp.ylabel("Output",fontsize=12)
mp.legend()
mp.grid()
```

OUTPUT :-

Activation Function



0.6 0.8 1.0